

Operational Safety and Cybersecurity Assurance of AI-Based Perception in Highly Automated Driving

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Abstract—In 2026 the United Nations adopted the first global framework for fully driverless automated driving systems that bases approval on a safety case and obliges the manufacturer to monitor operation and report field occurrences. For an AI-based perception component this operational evidence is indirect, because runtime provides no ground truth against which a detection can be compared. Monitors report predictive uncertainty, distribution shift, and cybersecurity events instead. ISO 21448 (safety of the intended functionality), ISO/PAS 8800 (AI safety), and ISO/SAE 21434 (cybersecurity) each define a concern-specific re-evaluation activity that uses this evidence. A runtime anomaly arrives without a concern label, because the same degraded detection can follow from a performance insufficiency or from an attack. No clause in these standards assigns such an anomaly to a concern, and none states how the concern-specific re-evaluations resolve into one judgment on the assurance argument. This paper states the gap at the clause level, separates it from the anomaly-detection problem existing work addresses, and specifies the requirements a resolution method must satisfy, leaving the method to future work.

Index Terms—assurance argument, safety case, cybersecurity, runtime monitoring, AI-based perception, automated driving

I. INTRODUCTION

This paper addresses assurance of AI-based perception in highly automated driving. The 2026 United Nations framework for fully driverless automated driving systems bases approval on a safety case and requires a safety management system, in-service monitoring and reporting, and a data storage system (WP.29) [1]. Operation-phase evidence thus informs the safety case throughout deployment. For an AI-based perception component this evidence is indirect. Unlike a planning function, whose output can be checked at runtime against traffic rules, a perception function has no ground truth to check a detection against. In operation it produces monitor outputs: predictive uncertainty and ensemble disagreement [2], distribution-shift indicators, and cybersecurity events [3]. Re-evaluation is activated by a change in monitor output, or by a software update, for which the framework provides [1].

An anomaly in this evidence is an abnormal monitor output, for example, a scene region where detections drop. It carries no concern label and does not show whether the cause falls to safety, AI safety, or cybersecurity. The same missing region follows from rain, sparse returns, or a rare object pose as a performance insufficiency under ISO 21448, the standard for

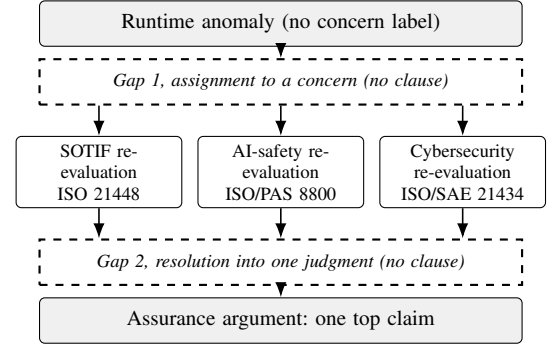


Fig. 1. A single runtime anomaly can activate any of the concern-specific re-evaluations (solid). Assignment to a concern (Gap 1) and resolution into one judgment (Gap 2) are dashed because no clause defines them.

the safety of the intended functionality (SOTIF), and from a spoofing or relay attack as a cybersecurity event under ISO/SAE 21434 [4]. These readings activate the concern-specific re-evaluations defined in separate standards (Fig. 1), yet no clause assigns the anomaly to one of them (the assignment gap), and no clause resolves their results into one judgment on whether the assurance argument still holds (the resolution gap). No published incident attributes a deployed automated-driving failure to this ambiguity. The motivation rests on the reporting duty, a demonstrated on-road attack [4], and the absence of a method to separate the two causes [5]. This paper takes the position that this resolution is the element of operational assurance the standards leave undefined, that it belongs at the assurance argument where the concern-specific activities meet, and that it is an in-service re-evaluation of the safety case, not an in-vehicle runtime function. The paper specifies the requirements a resolution method must satisfy.

II. RE-EVALUATION DEFINED BY EACH STANDARD

The pattern in Table I is uniform. Each activity is defined within one concern and produces a result for that concern alone. ISO/SAE 21434 differs. Its continual monitoring ends in risk treatment, and no clause re-evaluates the cybersecurity case [3]. The activities converge only at the assurance argument, where the SOTIF, AI-safety, and cybersecurity claims sit under one top claim, the top of the safety case [6], [7], [8].

TABLE I
OPERATION-PHASE RE-EVALUATION DEFINED BY EACH STANDARD, AND
THE ELEMENT EACH LEAVES UNDEFINED

Standard	Re-evaluation defined	Element not defined
ISO 21448 (SOTIF)	SOTIF argument on a field result (Clause 13.4)	Result-to-argument mapping
ISO/PAS 8800 (AI safety)	AI-safety argument; re-evaluate after change	Which evidence re-opens which element
ISO/SAE 21434 (cybersecurity)	Monitor, evaluate, treat	Re-evaluation of the case; change management scoped to requirements
None of these standards assigns an unlabeled anomaly to a concern, or resolves the concern-specific re-evaluations into one judgment on the assurance argument.		

III. ASSIGNMENT AND RESOLUTION OF THE ANOMALY

The anomaly has one form in the data and two causes, a performance insufficiency or an attack. A security survey confirms this: abnormal sensor readings can originate from an attack or from a non-malicious cause, and no method distinguishes the two from the readings alone. Multi-sensor cross-validation detects an anomaly but does not assign it to a cause [5]. Processing the anomaly within one concern presupposes an assignment that no clause performs. Without it, the re-evaluations can all act and produce results with no rule to combine them. Each can instead treat the anomaly as another's responsibility, so none acts. ISO 21448, in Clause 1 and Table 1, excludes cybersecurity threats and refers an attack to ISO/SAE 21434, while ISO/SAE 21434 does not address performance insufficiency [3], [6]. They can also reach opposed results, one keeping its concern's claim while another withdraws it.

Assignment alone would not close the gap. With every anomaly labeled, the concern-specific re-evaluations still meet at the top claim, where the evidence has different measurement scales, such as a calibrated probability, an ensemble disagreement value, a binary event indicator, or a risk rating, and no rule combines them. Operation adds conditions that design-time analysis does not face: asynchronous evidence, a component that changes under updates, and a reporting obligation that requires a current judgment on demand [1].

IV. RELATION TO EXISTING WORK

Machine-learning assurance guidance provides an AI argument structure not linked to the automotive standards' clauses [9], and dynamic safety cases stay within the safety concern [10]. Across both concerns, an argument-pattern method structures them at design time without defining how runtime evidence is interpreted [11], and a co-assurance framework leaves cross-concern operational evidence as an open problem [12]. A runtime method distinguishes attack from fault using branches already labeled attack or fault in a fault tree, not at the assurance argument [13]. None assigns a runtime monitor output across both concerns and then resolves the concern-specific re-evaluations into one judgment on the argument.

V. THE POSITION: REQUIREMENTS FOR A RESOLUTION METHOD

The concern-specific activities have no shared procedure or output below the argument [8]. A method placed below

that point reproduces the assignment problem within one concern. Each requirement follows from one feature of the gap and is necessary but not sufficient. **R1, heterogeneous evidence:** accept evidence with different measurement scales at one claim without imposing a common scale that no source standard defines. **R2, assignment:** contain an assignment step for an unlabeled anomaly, and treat the unassignable case as a defined outcome with a stated effect on the judgment. **R3, traceability:** produce a judgment traceable to the re-evaluation activities in ISO 21448, ISO/PAS 8800, and ISO/SAE 21434 and reportable under the in-service obligation [1]. **R4, modification:** state which evidence remains valid after a component update and which is re-opened, consistent with the modification provisions of ISO/PAS 8800 and ISO 26262 [14]. A candidate direction treats the anomaly as a defeater of the top claim, following eliminative argumentation [15]: assignment attaches it to a concern's sub-claim, resolution propagates its status to the top claim, an unresolved defeater is R2's residual outcome, and defeaters combine as claim status, not on a common scale, as R1 requires. Whether this fully satisfies R1–R4, and how to benchmark it, remain open.

VI. IMPLICATIONS FOR INDUSTRY AND STANDARDIZATION

Under WP.29 a manufacturer reports field occurrences against a safety case fixed at approval, with no defined procedure to turn a field anomaly into a current judgment on that case, on which continued operation depends [1]. The gap identifies what a revised ISO/PAS 8800 and ISO/SAE 21434 could add: an assignment step and a resolution step at the interface among the concern-specific activities. The same gap can arise wherever one component is assured against several concern-specific standards.

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